

Lab 03: Real-World Data Linear Regression

Least-squares fit of annual mean atmospheric CO₂ concentration versus year, using data from the NOAA Mauna Loa Observatory record.

Part A – Data

Source: NOAA Global Monitoring Laboratory, Keeling Curve / Mauna Loa Observatory CO₂ record.

URL: <https://gml.noaa.gov/ccgg/trends/> (data file:
https://gml.noaa.gov/webdata/ccgg/trends/co2/co2_annmean_mlo.csv)

Description: Since 1958, NOAA and the Scripps Institution of Oceanography have measured atmospheric carbon dioxide concentration at the Mauna Loa Observatory in Hawaii – the longest continuous record of atmospheric CO₂ in the world. The annual mean is the arithmetic mean of the twelve monthly average concentrations for that year. The 16 most recent complete years (2009–2024) are used here.

Variables: independent variable x = Year (calendar year); dependent variable y = annual mean atmospheric CO₂ concentration, in parts per million (ppm).

Year (x)	CO ₂ , ppm (y)	Year (x)	CO ₂ , ppm (y)
2009	387.64	2017	406.76
2010	390.10	2018	408.72
2011	391.85	2019	411.65
2012	394.06	2020	414.21
2013	396.74	2021	416.41
2014	398.81	2022	418.53
2015	401.01	2023	421.08
2016	404.41	2024	424.61

Part B – Python Script

Complete script (also submitted separately as **co2_regression.py**). The slope and intercept are computed directly from the least-squares normal equations; numpy is used only for elementary sums.

```
"""
Lab 03: Real-World Data Linear Regression
Numerical Methods, Section 3D

Dataset: Annual mean atmospheric CO2 concentration at Mauna Loa Observatory
Source: NOAA Global Monitoring Laboratory (Keeling Curve)
        https://gml.noaa.gov/ccgg/trends/ (data file: co2_annmean_mlo.csv)

x = Year
y = Annual mean CO2 concentration (ppm)

The regression is performed "by hand" using the least-squares normal
equations for a straight line, exactly as derived in class:

    a1 = ( n*sum(x*y) - sum(x)*sum(y) ) / ( n*sum(x^2) - (sum(x))^2 )
    a0 = ybar - a1*xbar

No numpy.polyfit / scipy curve-fitting routine is used for the fit itself;
numpy is only used for basic array/sum bookkeeping.
"""

import numpy as np
import matplotlib
matplotlib.use("Agg")
import matplotlib.pyplot as plt

# -----
# 1. The data (16 paired observations, 2009-2024)
# -----
year = np.array([2009, 2010, 2011, 2012, 2013, 2014, 2015, 2016,
                 2017, 2018, 2019, 2020, 2021, 2022, 2023, 2024])

co2_ppm = np.array([387.64, 390.10, 391.85, 394.06, 396.74, 398.81,
                   401.01, 404.41, 406.76, 408.72, 411.65, 414.21,
                   416.41, 418.53, 421.08, 424.61])

x = year.astype(float)
y = co2_ppm.astype(float)
n = len(x)

# -----
# 2. Least-squares straight-line fit: y = a0 + a1*x
# -----
sum_x = np.sum(x)
sum_y = np.sum(y)
sum_xy = np.sum(x * y)
sum_x2 = np.sum(x ** 2)

a1 = (n * sum_xy - sum_x * sum_y) / (n * sum_x2 - sum_x ** 2)
a0 = (sum_y - a1 * sum_x) / n

print(f"Number of observations, n = {n}")
print(f"Slope, a1 = {a1:.4f} ppm/year")
print(f"Intercept, a0 = {a0:.4f} ppm")
print(f"Regression equation: y = {a0:.3f} + {a1:.4f} x")

# -----
# 3. Goodness of fit: Sr (SSE), St (SST), r^2, standard error sy/x
# -----
y_pred = a0 + a1 * x
residuals = y - y_pred

St = np.sum((y - np.mean(y)) ** 2) # total sum of squares
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Sr = np.sum(residuals ** 2)          # sum of squares of residuals (SSE)
r2 = 1 - Sr / St                     # coefficient of determination
syx = np.sqrt(Sr / (n - 2))          # standard error of the estimate

print(f"\nSt (total sum of squares)    = {St:.4f}")
print(f"Sr (sum of squares of resid.) = {Sr:.4f}")
print(f"r^2 (coefficient of determin.) = {r2:.6f}")
print(f"r (correlation coefficient)    = {np.sqrt(r2):.6f}")
print(f"s_y/x (standard error)         = {syx:.4f} ppm")

# -----
# 4. Prediction for a year NOT in the dataset
# -----
x_new = 2030
y_new = a0 + a1 * x_new
print(f"\nPrediction: for x = {x_new}, predicted y = {y_new:.2f} ppm")

# -----
# 5. Plot 1: data + fitted line
# -----
plt.figure(figsize=(7, 4.5))
plt.scatter(x, y, color="#1f4e79", label="Observed annual mean CO2", zorder=3)
x_line = np.linspace(x.min() - 1, x.max() + 1, 100)
plt.plot(x_line, a0 + a1 * x_line, color="#c8963e", linewidth=2,
         label=f"Fit: y = {a0:.2f} + {a1:.3f}x")
plt.xlabel("Year")
plt.ylabel("Annual mean CO2 concentration (ppm)")
plt.title("Mauna Loa Annual Mean Atmospheric CO2, 2009-2024")
plt.legend()
plt.grid(alpha=0.3)
plt.tight_layout()
plt.savefig("fit_plot.png", dpi=200)
plt.close()

# -----
# 6. Plot 2: residual plot
# -----
plt.figure(figsize=(7, 4.5))
plt.axhline(0, color="black", linewidth=1)
plt.scatter(x, residuals, color="#1f4e79", zorder=3)
plt.xlabel("Year")
plt.ylabel("Residual, y - y_predicted (ppm)")
plt.title("Residual Plot")
plt.grid(alpha=0.3)
plt.tight_layout()
plt.savefig("residual_plot.png", dpi=200)
plt.close()

print("\nSaved fit_plot.png and residual_plot.png")

```

Part C – Regression Results

Regression equation with fitted values substituted in:

$$y = -4528.756 + 2.4469 x$$

Quantity	Symbol	Value
Number of observations	n	16
Slope	a1	2.4469 ppm / year
Intercept	a0	-4528.756 ppm
Sum of squares of residuals (SSE)	Sr	2.7497 ppm2
Total sum of squares	St	2038.4334 ppm2
Coefficient of determination	r2	0.998651
Correlation coefficient	r	0.999325
Standard error of the estimate	sy/x	0.4432 ppm

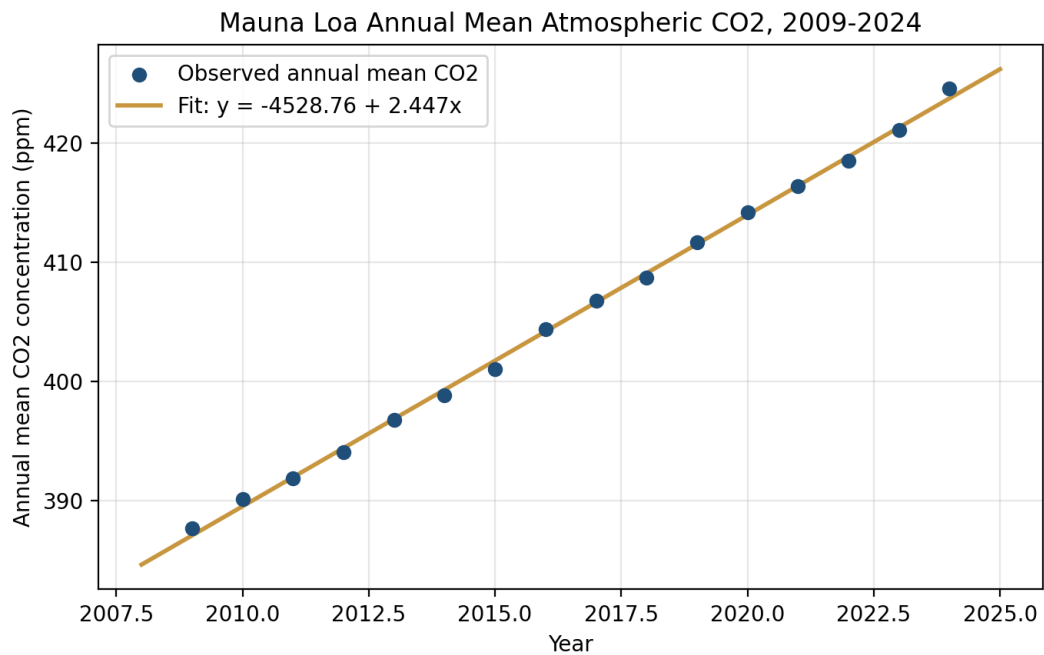


Figure 1. Annual mean CO₂ concentration at Mauna Loa (2009–2024) with the least-squares fitted line.

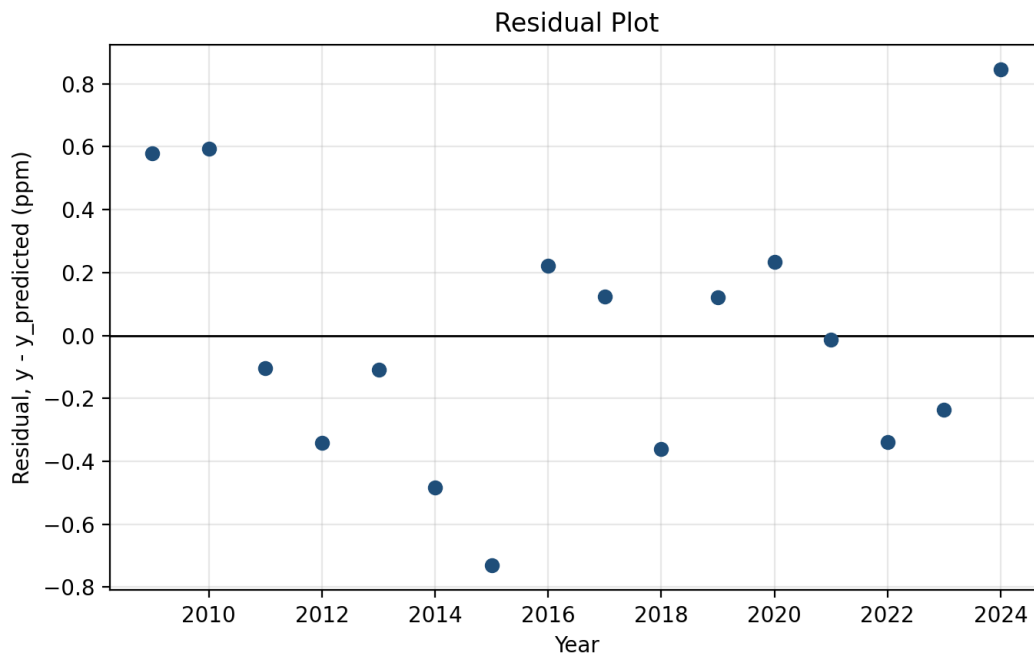


Figure 2. Residuals (observed – predicted) versus year.

Prediction

For $x = 2030$ (a year outside the fitted data range):

$$y = -4528.756 + 2.4469(2030) = 438.44 \text{ ppm}$$

This prediction is meaningful as a short-range extrapolation: the CO₂ record has followed a very steady, almost perfectly linear rise over the past 16 years, and 2030 is only six years beyond the last observed point, so the physical process (fossil-fuel driven CO₂ accumulation) is not expected to change its short-term linear behavior in that time. Extrapolating much further into the future would be riskier, since the true long-term trend is actually slightly accelerating (curving upward), not perfectly linear.

Interpretation

Slope and intercept: The slope $a_1 = 2.4469$ ppm/year means that, over 2009–2024, the annual mean atmospheric CO₂ concentration increased by about 2.45 ppm every year on average. The intercept $a_0 = -4528.76$ ppm is the value the line would predict at year $x = 0$; it has no direct physical meaning by itself (year 0 is far outside the data range) and should be treated only as a fitting parameter needed to position the line correctly.

Goodness of fit: $r^2 = 0.9987$ means that about 99.9% of the year-to-year variation in CO₂ concentration is explained by the linear trend with year. The standard error of the estimate, $sy/x = 0.44$ ppm, tells us that a typical observation deviates from the fitted line by less than half a part per million – a very tight fit given that CO₂ rose from about 388 to 425 ppm over the period.

Residuals: The residual plot shows no obvious funnel shape or strong curvature, but the points do trace a shallow, smile-shaped (U-shaped) pattern: residuals are slightly negative in the middle years and slightly positive at both ends. This is the signature of a small amount of upward curvature (acceleration) in the true CO₂ trend that a straight line cannot capture. In other words, a straight line is an excellent local

approximation over this 16-year window, but a quadratic model would likely fit the data slightly better.